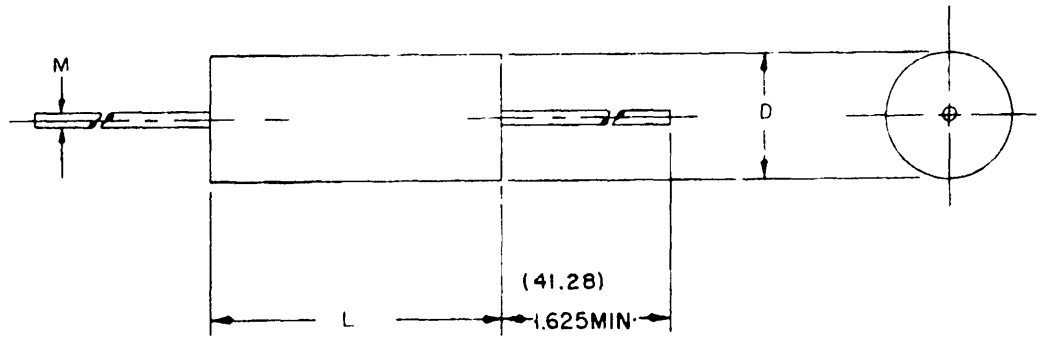
MIL-C-55514/1A
22 July 1970
SUPERSEDING
MIL-C-55514/1 (EL) IN PART
29 September 1967
MIL-C-27287/1 (USAF)
28 June 1960

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY, STYLE CF02

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-55514.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



NOTES:
1. Dimensions are in inches.
2. Leads shall be of solid wire and located on centerline within ±.062 (1.57) but not less than .03 (.76) from edge of capacitor.
3. Millimeters are in parentheses.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style CF02 capacitors.

FSC 5910

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TABLE I. Style CF02 capacitor characteristics

CHARACTERISTIC M CAPACITORS						
Type designation 1/	Capacitance	Rated voltage (at 85° C)	Capacitance tolerance available	Dimensions (in inches)		
				D 2/	L ±.062(1.57)	M $^{+.005(.13)}_{-.003(.08)}$ 3/
	µF	volts, dc				
CF02AMB682--	.0068	100	G, J, K	.172 (4.37)	.438(11.13)	.025(.64)
CF02AMB822--	.0082	100		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMB103--	.010	100		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMB123--	.012	100		.172 (4.37)	.688(17.48)	.025(.64)
CF02AMB153--	.015	100		.172 (4.37)	.688(17.48)	.025(.64)
CF02AMB183--	.018	100		.188 (4.78)	.688(17.48)	.025(.64)
CF02AMB223--	.022	100		.188 (4.78)	.688(17.48)	.025(.64)
CF02AMB273--	.027	100		.203 (5.16)	.688(17.48)	.025(.64)
CF02AMB333--	.033	100		.203 (5.16)	.688(17.48)	.025(.64)
CF02AMB393--	.039	100		.250 (6.35)	.688(17.48)	.025(.64)
CF02AMB473--	.047	100		.250 (6.35)	.688(17.48)	.025(.64)
CF02AMB563--	.056	100		.250 (6.35)	.812(20.62)	.025(.64)
CF02AMB683--	.068	100		.250 (6.35)	.812(20.62)	.025(.64)
CF02AMB823--	.082	100		.297 (7.54)	.812(20.62)	.025(.64)
CF02AMB104--	.10	100		.297 (7.54)	.812(20.62)	.025(.64)
CF02AMB124--	.12	100		.360 (9.14)	.812(20.62)	.032(.81)
CF02AMB154--	.15	100		.360 (9.14)	.812(20.62)	.032(.81)
CF02AMB184--	.18	100		.329 (8.36)	1.188(30.18)	.025(.64)
CF02AMB224--	.22	100		.329 (8.36)	1.188(30.18)	.025(.64)
CF02AMB274--	.27	100		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AMB334--	.33	100		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AMB394--	.39	100		.485(12.32)	1.188(30.18)	.032(.81)
CF02AMB474--	.47	100		.485(12.32)	1.188(30.18)	.032(.81)
CF02AMB564--	.56	100		.500(12.70)	1.562(39.67)	.032(.81)
CF02AMB684--	.68	100		.500(12.70)	1.562(39.67)	.032(.81)
CF02AMB824--	.82	100		.548(13.92)	1.562(39.67)	.032(.81)
CF02AMB105--	1.00	100		.548(13.92)	1.562(39.67)	.032(.81)
CF02AMC102--	.0010	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC122--	.0012	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC152--	.0015	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC182--	.0018	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC222--	.0022	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC272--	.0027	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC332--	.0033	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC392--	.0039	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC472--	.0047	200		.172 (4.37)	.438(11.13)	.025(.64)
CF02AMC562--	.0056	200		.172 (4.37)	.688(17.48)	.025(.64)
CF02AMC682--	.0068	200		.172 (4.37)	.688(17.48)	.025(.64)
CF02AMC822--	.0082	200		.188 (4.78)	.688(17.48)	.025(.64)
CF02AMC103--	.010	200		.188 (4.78)	.688(17.48)	.025(.64)
CF02AMC123--	.012	200		.203 (5.16)	.688(17.48)	.025(.64)
CF02AMC153--	.015	200		.203 (5.16)	.688(17.48)	.025(.64)
CF02AMC183--	.018	200		.234 (5.94)	.688(17.48)	.025(.64)
CF02AMC223--	.022	200		.234 (5.94)	.688(17.48)	.025(.64)
CF02AMC273--	.027	200		.250 (6.35)	.812(20.62)	.025(.64)
CF02AMC333--	.033	200		.250 (6.35)	.812(20.62)	.025(.64)
CF02AMC393--	.039	200		.266 (6.76)	.812(20.62)	.025(.64)
CF02AMC473--	.047	200		.266 (6.76)	.812(20.62)	.025(.64)
CF02AMC563--	.056	200	G, J, K	.297 (7.54)	.812(20.62)	.025(.64)

See footnotes at end of table.

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TABLE I. Style CF02 capacitor characteristics (Cont'd)

CHARACTERISTIC M CAPACITORS						
Type designation 1/	Capacitance	Rated voltage (at 85° C)	Capacitance tolerance available	Dimensions (in inches)		
				D 2/	L ±.062(1.57)	M $^{+.005(.13)}_{-.003(.08)}$ 3/
	μF	volts, dc				
CF02AMC683--	.068	200	G, J, K	.297 (7.54)	.812(20.62)	.025(.64)
CF02AMC823--	.082	200		.329 (8.36)	1.062(26.97)	.025(.64)
CF02AMC104--	.10	200		.329 (8.36)	1.062(26.97)	.025(.64)
CF02AMC124--	.12	200		.390 (9.91)	1.062(26.97)	.032(.81)
CF02AMC154--	.15	200		.390 (9.91)	1.062(26.97)	.032(.81)
CF02AMC184--	.18	200		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AMC224--	.22	200		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AMC274--	.27	200		.500(12.70)	1.188(30.18)	.032(.81)
CF02AMC334--	.33	200		.500(12.70)	1.188(30.18)	.032(.81)
CF02AMC394--	.39	200		.516(13.11)	1.562(39.67)	.032(.81)
CF02AMC474--	.47	200		.516(13.11)	1.562(39.67)	.032(.81)
CF02AMC564--	.56	200		.630(16.00)	1.688(42.48)	.032(.81)
CF02AMC684--	.68	200		.630(16.00)	1.688(42.48)	.032(.81)
CF02AMC824--	.82	200		.695(17.65)	1.875(47.63)	.032(.81)
CF02AMC105--	1.00	200		.695(17.65)	1.875(47.63)	.032(.81)
CF02AME182--	.0018	400		.172 (4.37)	.688(17.48)	.025(.64)
CF02AME222--	.0022	400		.172 (4.37)	.688(17.48)	.025(.64)
CF02AME272--	.0027	400		.188 (4.78)	.688(17.48)	.025(.64)
CF02AME332--	.0033	400		.188 (4.78)	.688(17.48)	.025(.64)
CF02AME392--	.0039	400		.203 (5.16)	.688(17.48)	.025(.64)
CF02AME472--	.0047	400		.203 (5.16)	.688(17.48)	.025(.64)
CF02AME562--	.0056	400		.250 (6.35)	.688(17.48)	.025(.64)
CF02AME682--	.0068	400		.250 (6.35)	.688(17.48)	.025(.64)
CF02AME822--	.0082	400		.250 (6.35)	.812(20.62)	.025(.64)
CF02AME103--	.010	400		.250 (6.35)	.812(20.62)	.025(.64)
CF02AME123--	.012	400		.266 (6.76)	.812(20.62)	.025(.64)
CF02AME153--	.015	400		.266 (6.76)	.812(20.62)	.025(.64)
CF02AME183--	.018	400		.313 (7.95)	.812(20.62)	.025(.64)
CF02AME223--	.022	400		.313 (7.95)	.812(20.62)	.025(.64)
CF02AME273--	.027	400		.360 (9.14)	.812(20.62)	.032(.81)
CF02AME333--	.033	400		.360 (9.14)	.812(20.62)	.032(.81)
CF02AME393--	.039	400		.360 (9.14)	1.062(26.97)	.032(.81)
CF02AME473--	.047	400		.360 (9.14)	1.062(26.97)	.032(.81)
CF02AME563--	.056	400		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AME683--	.068	400		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AME823--	.082	400		.485(12.32)	1.188(30.18)	.032(.81)
CF02AME104--	.10	400		.485(12.32)	1.188(30.18)	.032(.81)
CF02AME124--	.12	400		.500(12.70)	1.562(39.67)	.032(.81)
CF02AME154--	.15	400		.500(12.70)	1.562(39.67)	.032(.81)
CF02AME184--	.18	400		.586(14.88)	1.562(39.67)	.032(.81)
CF02AME224--	.22	400		.586(14.88)	1.562(39.67)	.032(.81)
CF02AME274--	.27	400		.681(17.30)	1.688(42.48)	.032(.81)
CF02AME334--	.33	400		.681(17.30)	1.688(42.48)	.032(.81)
CF02AME394--	.39	400		.750(19.05)	1.875(47.63)	.032(.81)
CF02AME474--	.47	400	G, J, K	.750(19.05)	1.875(47.63)	.032(.81)

See footnotes at end of table.

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TABLE I. Style CF02 capacitor characteristics (Cont'd)

Type designation 1/	Capacitance	Rated voltage (at 85° C)	Capacitance tolerance available	Dimensions (in inches)		
				D 2/	L ± .062(1.57)	M $^{+.005(.13)}_{-.003(.08)}$ 3/
	µF	volts, dc				
CHARACTERISTIC M CAPACITORS						
CF02AME564--	.56	400	G, J, K	.940(23.88)	2.125(53.98)	.032(.81)
CF02AME684--	.68	400		1.020(25.91)	2.375(60.33)	.032(.81)
CF02AME824--	.82	400		1.150(29.21)	2.375(60.33)	.032(.81)
CF02AME105--	1.00	400		1.150(29.21)	2.625(66.68)	.032(.81)
CF02AMF102--	.0010	600		.172 (4.37)	.688(17.48)	.025(.64)
CF02AMF122--	.0012	600		.172 (4.37)	.688(17.48)	.025(.64)
CF02AMF152--	.0015	600		.172 (4.37)	.688(17.48)	.025(.64)
CF02AMF182--	.0018	600		.188 (4.78)	.688(17.48)	.025(.64)
CF02AMF222--	.0022	600		.188 (4.78)	.688(17.48)	.025(.64)
CF02AMF272--	.0027	600		.203 (5.16)	.688(17.48)	.025(.64)
CF02AMF332--	.0033	600		.203 (5.16)	.688(17.48)	.025(.64)
CF02AMF392--	.0039	600		.250 (6.35)	.688(17.48)	.025(.64)
CF02AMF472--	.0047	600		.250 (6.35)	.688(17.48)	.025(.64)
CF02AMF562--	.0056	600		.250 (6.35)	.812(20.62)	.025(.64)
CF02AMF682--	.0068	600		.250 (6.35)	.812(20.62)	.025(.64)
CF02AMF822--	.0082	600		.266 (6.76)	.812(20.62)	.025(.64)
CF02AMF103--	.010	600		.266 (6.76)	.812(20.62)	.025(.64)
CF02AMF123--	.012	600		.313 (7.95)	.812(20.62)	.025(.64)
CF02AMF153--	.015	600		.313 (7.95)	.812(20.62)	.025(.64)
CF02AMF183--	.018	600		.360 (9.14)	.812(20.62)	.032(.81)
CF02AMF223--	.022	600		.360 (9.14)	.812(20.62)	.032(.81)
CF02AMF273--	.027	600		.360 (9.14)	1.062(26.97)	.032(.81)
CF02AMF333--	.033	600		.360 (9.14)	1.062(26.97)	.032(.81)
CF02AMF393--	.039	600		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AMF473--	.047	600		.390 (9.91)	1.188(30.18)	.032(.81)
CF02AMF563--	.056	600		.485(12.32)	1.188(30.18)	.032(.81)
CF02AMF683--	.068	600		.485(12.32)	1.188(30.18)	.032(.81)
CF02AMF823--	.082	600		.500(12.70)	1.562(39.67)	.032(.81)
CF02AMF104--	.10	600	G, J, K	.500(12.70)	1.562(39.67)	.032(.81)
CHARACTERISTIC Q CAPACITORS						
CF02AQB102--	.0010	100	F, G, J, K	.231 (5.87)	.453(11.51)	.025(.64)
CF02AQB122--	.0012	100		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQB152--	.0015	100		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQB182--	.0018	100		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQB222--	.0022	100		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQB272--	.0027	100		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQB332--	.0033	100		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQB392--	.0039	100		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQB472--	.0047	100		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQB562--	.0056	100		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQB682--	.0068	100		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQB822--	.0082	100		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQB103--	.010	100		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQB123--	.012	100		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQB153--	.015	100		.312 (7.92)	.453(11.51)	.025(.64)
CF02AQB183--	.018	100	F, G, J, K	.328 (8.33)	.453(11.51)	.025(.64)

See footnotes at end of table.

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TABLE I. Style CF02 capacitor characteristics (Cont'd)

CHARACTERISTIC Q CAPACITORS						
Type designation 1/	Capacitance	Rated voltage (at 85° C)	Capacitance tolerance available	Dimensions (in inches)		
				D 2/	L ±.062(1.57)	M +.005(.13) -.003(.08) 3/
	µF	volts, dc				
CF02AQB223--	.022	100	F, G, J, K	.266 (6.76)	.750(19.05)	.025(.64)
CF02AQB273--	.027	100		.281 (7.14)	.750(19.05)	.025(.64)
CF02AQB333--	.033	100		.297 (7.54)	.750(19.05)	.025(.64)
CF02AQB393--	.039	100		.312 (7.92)	.750(19.05)	.025(.64)
CF02AQB473--	.047	100		.328 (8.33)	.750(19.05)	.025(.64)
CF02AQB563--	.056	100		.359 (9.12)	.750(19.05)	.032(.81)
CF02AQB683--	.068	100		.391 (9.93)	.750(19.05)	.032(.81)
CF02AQB823--	.082	100		.359 (9.12)	.875(22.23)	.032(.81)
CF02AQB104--	.100	100		.406(10.31)	.875(22.23)	.032(.81)
CF02AQB124--	.12	100		.422(10.72)	.875(22.23)	.032(.81)
CF02AQB154--	.15	100		.468(11.89)	.875(22.23)	.032(.81)
CF02AQB184--	.18	100		.406(10.31)	1.219(30.96)	.032(.81)
CF02AQB224--	.22	100		.453(11.51)	1.219(30.96)	.032(.81)
CF02AQB274--	.27	100		.484(12.29)	1.219(30.96)	.032(.81)
CF02AQB334--	.33	100		.516(13.11)	1.219(30.96)	.032(.81)
CF02AQB394--	.39	100		.549(13.94)	1.219(30.96)	.032(.81)
CF02AQB474--	.47	100		.625(15.88)	1.219(30.96)	.032(.81)
CF02AQB564--	.56	100		.688(17.48)	1.219(30.96)	.032(.81)
CF02AQB684--	.68	100		.641(16.29)	1.594(40.49)	.032(.81)
CF02AQB824--	.82	100		.688(17.48)	1.594(40.49)	.032(.81)
CF02AQB105--	1.00	100		.719(18.26)	1.594(40.49)	.032(.81)
CF02AQC102--	.0010	200		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQC122--	.0012	200		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQC152--	.0015	200		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQC182--	.0018	200		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQC222--	.0022	200		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQC272--	.0027	200		.231 (5.87)	.453(11.51)	.025(.64)
CF02AQC332--	.0033	200		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQC392--	.0039	200		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQC472--	.0047	200		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQC562--	.0056	200		.281 (7.14)	.453(11.51)	.025(.64)
CF02AQC682--	.0068	200		.312 (7.92)	.453(11.51)	.025(.64)
CF02AQC822--	.0082	200		.231 (5.87)	.750(19.05)	.025(.64)
CF02AQC103--	.010	200		.250 (6.35)	.750(19.05)	.025(.64)
CF02AQC123--	.012	200		.266 (6.76)	.750(19.05)	.025(.64)
CF02AQC153--	.015	200		.281 (7.14)	.750(19.05)	.025(.64)
CF02AQC183--	.018	200		.312 (7.92)	.750(19.05)	.025(.64)
CF02AQC223--	.022	200		.328 (8.33)	.750(19.05)	.025(.64)
CF02AQC273--	.027	200		.359 (9.12)	.750(19.05)	.032(.81)
CF02AQC333--	.033	200		.391 (9.93)	.750(19.05)	.032(.81)
CF02AQC393--	.039	200		.359 (9.12)	.875(22.23)	.032(.81)
CF02AQC473--	.047	200		.391 (9.93)	.875(22.23)	.032(.81)
CF02AQC563--	.056	200		.422(10.72)	.875(22.23)	.032(.81)
CF02AQC683--	.068	200		.453(11.51)	.875(22.23)	.032(.81)
CF02AQC823--	.082	200		.500(12.70)	.875(22.23)	.032(.81)
CF02AQC104--	.10	200		.438(11.13)	1.219(30.96)	.032(.81)
CF02AQC124--	.12	200		.469(11.91)	1.219(30.96)	.032(.81)
CF02AQC154--	.15	200	F, G, J, K	.516(13.11)	1.219(30.96)	.032(.81)

See footnotes at end of table.

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TABLE I. Style CF02 capacitor characteristics (Cont'd)

CHARACTERISTIC Q CAPACITORS						
Type designation 1/	Capacitance	Rated voltage (at 85° C)	Capacitance tolerance available	Dimensions (in inches)		
				D 2/	L ±.062(1.57)	M $^{+.005(.13)}_{-.003(.08)}$ 3/
	μF	volts, dc				
CF02AQC184--	.18	200	F, G, J, K	.547(13.89)	1.219(30.96)	.032(.81)
CF02AQC224--	.22	200		.625(15.88)	1.219(30.96)	.032(.81)
CF02AQC274--	.27	200		.594(15.09)	1.594(40.49)	.032(.81)
CF02AQC334--	.33	200		.641(16.29)	1.594(40.49)	.032(.81)
CF02AQC394--	.39	200		.688(17.48)	1.594(40.49)	.032(.81)
CF02AQC474--	.47	200		.734(18.64)	1.594(40.49)	.032(.81)
CF02AQC564--	.56	200		.797(20.24)	1.594(40.49)	.032(.81)
CF02AQC684--	.68	200		.859(21.82)	1.594(40.49)	.032(.81)
CF02AQC824--	.82	200		.859(21.82)	1.844(46.84)	.032(.81)
CF02AQC105--	1.00	200	F, G, J, K	.938(23.83)	1.844(46.84)	.032(.81)

1/ The complete type designation will include additional symbols to indicate capacitance tolerance, and failure rate level (M, P, R, or S).

2/ Tolerances for dimension "D" are ±.031 (0.250 inch diameter and below); ±.046 (0.251 through 0.499 inch diameter); and ±.062 (0.500 inch diameter and above).

3/ Comparable AWG sizes are No. 22 (.025 inch) and No. 20 (.032 inch).

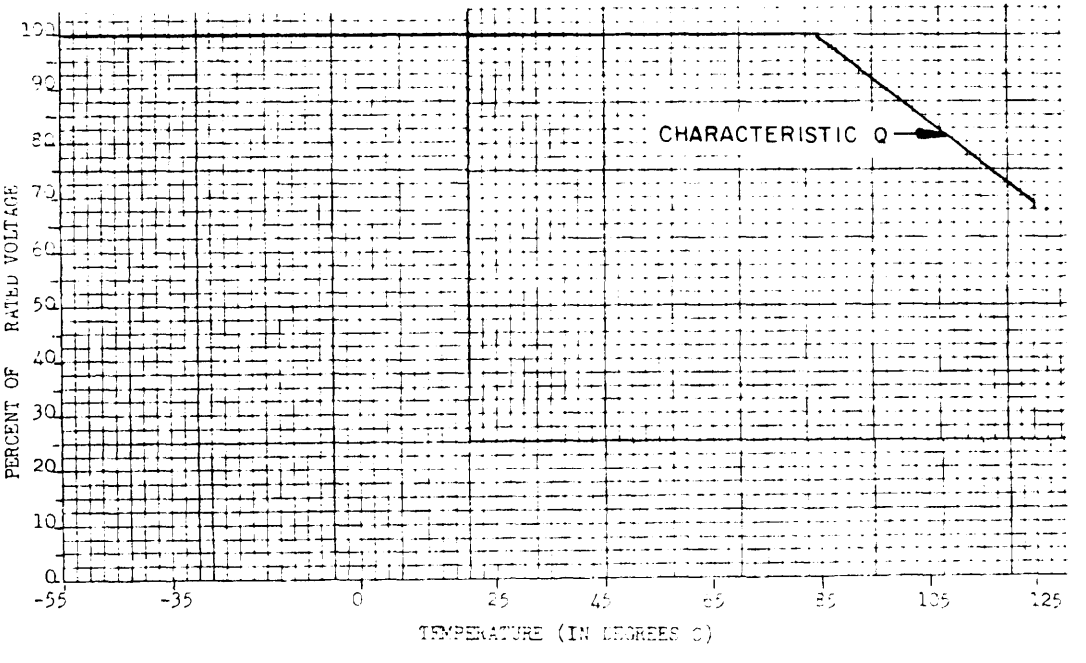


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table I.
Terminals - Axial wire-lead (A).
Characteristic - M or Q, as shown in table I.

Rated voltage - 100v (B), 200v (C), 400v (E) or 600v (F). When properly voltage derated (see figure 2), characteristic Q units will operate to 125°C.
Capacitance (nom) - See table I.
Capacitance tolerance - See table I.
Failure rate level - M, P, R, or S.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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